RANGAMATI SCIENCE AND TECHNOLOGY UNIVERSITY

Department of Computer Science and Engineering

4th Year 1st Semester BSc (Engr.) Final Examination - 2023

Session: 2019-2020

Course: Computer Networking; Code: CSE 4102

Duration: 03 Hours; Full Marks: 60

NB:	
1.	Answer any **FOUR (4)** questions out of **SIX (6)** questions.
2.	Figures in the right margin indicate marks(15*4=60).
3.	All parts of a question must be answered serially

1. a) List the layers of the TCP/IP model. Evaluate the trade-offs between the OSI model's abstraction and TCP/IP's real-world implementation. Suppose data corruption happens which layer is responsible for detection and recovery? Explain with an example. 06

   b) Explain the roles of a Switch, a Hub, and a MAC address in computer networking. 04

   c) Describe Multiplexing and De-multiplexing in terms of Transport Layer. 05

2. a) Justify the statement: "An IP address is network-specific rather than device-specific." 04

   b) Define routing. Analyze how switches reduce collisions in Ethernet LANs. In a campus network, justify whether routers or switches should be used at each level (core, distribution, access). 07

   c) Define Subnetting and Subnet Mask. Explain how a Subnet Mask works and how CIDR assists in subnetting with appropriate figure in your explanation. 04

3. a) Explain how a TCP connection is established between a client and a server. Describe each step in the process. 05

   b) Describe the format of IP addresses. Compare IPv4 and IPv6 addressing, and explain why IPv6 is necessary despite the existence of IPv4. 05

   c) Explain how the TCP/IP protocols work in terms of data transmission. 05

4. a) What is a network protocol? Compare the use of HTTP vs HTTPS in terms of security and performance. Suppose an online video service must choose between TCP and UDP which should it use and why? 06

   b) Explain ARP and how it functions, including the use of the ARP table. 05

   c) Describe the steps of a basic routing protocol. Illustrate your answer with a diagram. 04

5. a) Explain what a Firewall is, why it is necessary, and how it works in computer networks. 05

   b) Describe the different types of routing gateway protocols 05

   c) Critically analyze the role of authentication and access control in securing network systems. 03

d) Calculate the number of links required in a fully connected mesh with 8 devices. 02

6. a) What are the various purposes of DNS? Explain how it helps clients access websites without worrying about changes in IP addresses. 05

b) Explain the processes of encapsulation and de-encapsulation of a message across the different layers of a network. 05

c) What are the five primary types of DNS servers? Explain how caching and recursive servers work. 05

RANGAMATI SCIENCE AND TECHNOLOGY UNIVERSITY

Department of Computer Science and Engineering

4th Year 1st Semester BSc (Engg.) Final Examination - 2023

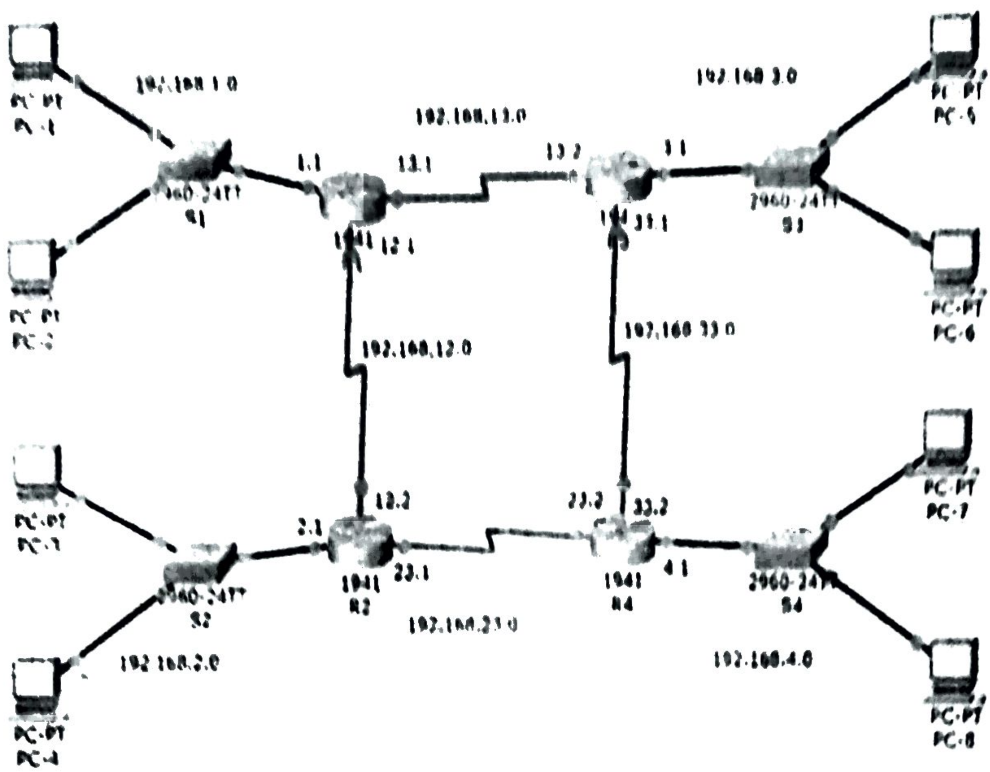
Session: 2019-2020

Course: Computer Networking Lab, Code: CSE-4104

Duration: 03 Hours. | Full Marks: 60

Implement the given network architecture using CISCO Packet Tracer on your laptop or lab PC. In your exam script, you must:

1. Draw the complete network topology
2. Write the step by step procedure for transferring data from PC-4 to PC-8.



Rangamati Science and Technology University

Department of Computer Science and Engineering

4th Year 1st Semester B.Sc. (Engg.) Final Exam-2023

Course Code: CSE-4104; **Session:** 2019-2020

Course Title: Computer Graphics

Time: 3 Hours **Marks:** 60

NB:
1. Answer any **FOUR (4)** questions out of **SIX (6)** questions.
2. Figures in the right margin indicate marks (15*4=60).
3. All parts of a question must be answered serially.
4. Keep your answer script clean and free from overwriting

1. (a) Define computer graphics. Mention the basic components of computer graphics. 3

   (b) Describe the working principle of cathode ray tube-CRT with its fundamental units. 4

   (c) Explain: 4

   i. RGB and CMY color model

   ii. Scaling and reflection with examples.

   (d) Define and plot the successive generations of the C Curve & Koch curve at level 4 of a straight line. 4

2. (a) Write the algorithm of midpoint ellipse algorithm step by step. 5

   (b) What steps are required to scan converting a circle using Bresenham's algorithm. 5

   (c) Use pseudo-code to describe the steps that are required to plot a line whose slope is between 45^0 and -45^0 where $|m| > 1$: using the slope intercept equation. 5

3. (a) State the names of different types of transformations. Mention your opinion on why it is important. 3

   (b) For the given matrix first apply a matrix of 60^0 angle about the Y-axis followed by the rotation of 60^0 about X-axis and now determined the resultant matrix. 4

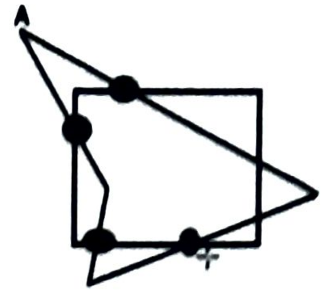
$$\begin{bmatrix} 2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 0 \\ 4 & 0 & 1 & 0 \\ 0 & 3 & 6 & 1 \end{bmatrix}$$

   (c) Evaluate the formula of translation vector (tx, ty), geometric and coordinate rotation in 2D & 3D transformation. 4

   (d) What is viewing and clipping? Express window to viewport mapping in the form of 2D coordinates. 4

4. (a) What is meant by shading? Classify different types of shading. 3

   (b) Using Sutherland Hogeman Polygon Clipping algorithm, clip the following polygon against each edge of window, show new set of vertices and write down the algorithm. 8

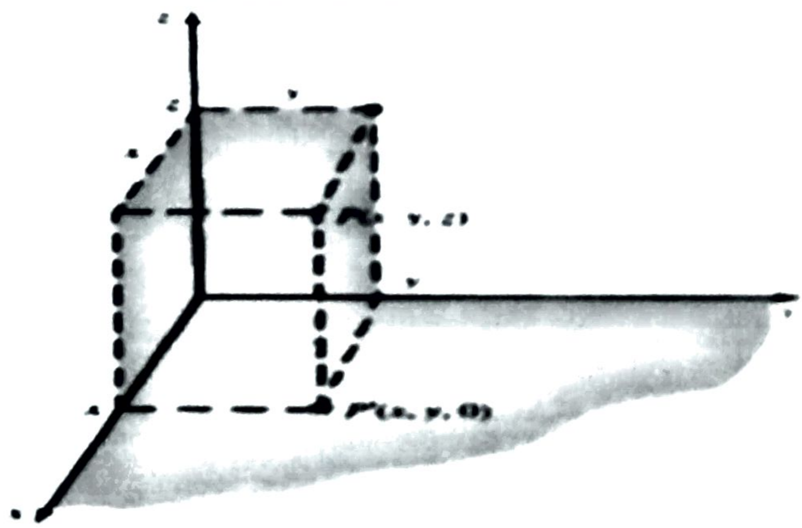


(c) A rectangular parallelepiped has its length as 3 units, 2 units, and 1 unit on x,y,z axis respectively, perform rotation by 90^0 clockwise about x-axis. 4

5. a) Show the taxonomy of projection in graphics. 3

b) Consider a unit cube **V** projected onto the xy plane given below. Draw the projected image using the standard perspective transformation with **(i) d=1** and **(ii) d=10,** where **d** is the distance from the view plane. 8

$$V = ABCDEFGH = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$



c) Define spline curve. Find the matrix for mirror reflection with respect to the plane passing through the origin and having a normal vector whose direction is N = I+J+K. 4

6. (a) Describe the line clipping process implementing the Cohen-Sutherland algorithm. 6

(b) Find the normalization transformation N which uses the rectangle A(1,1), B(5,3), C(4,5), D(0,3) as a window given at figure a, and the normalized device screen as a viewport given figure b. 5

(c) Show the window-to-viewport coordinate transformation. 4

Rangamati Science and Technology University

Department of Computer Science and Engineering

4th Year 1st Semester B.Sc. (Engg.) Final Exam 2023

Course Code: CSE-5105, **Session:** 2019-2020

Course Title: Natural Language Processing

Time: 3 Hours **Marks:** 60

NB	1	Answer any **FOUR (4)** questions out of **SIX (6)** questions.
	2	Figures in the right margin indicate marks(15*4=60)
	3	All parts of a question must be answered serially.

1.	(a)	Define and explain the core components of Natural Language Processing (NLP) Use the sentence below to demonstrate any two components. "The quick brown fox jumps over the lazy dog."	5
	(b)	Discuss the role of text preprocessing in NLP. Consider tokenization, stopword removal, stemming, lemmatization, and lowercasing. ii) How can each step affect model performance in tasks like sentiment analysis, spam detection, or topic modeling? iii) Give examples where excessive preprocessing might harm results.	6
	(c)	Differentiate between high-resource and low-resource languages in NLP. Provide one example of each and discuss the key challenges in low-resource NLP processing.	4
2.	(a)	Hate speech detection is a common NLP task where text must be labeled as Hate Speech, Offensive but Not Hate Speech, or Neutral. i) Describe two methods of data annotation for hate speech classification (e.g., manual annotation, crowdsourcing, semi-automatic annotation). Explain their advantages and limitations. ii) Outline three essential guidelines for annotators to maintain consistency and fairness in hate speech labeling. Why are these guidelines important for reducing bias? iii) Consider the following example sentences and annotate them according to the three classes mentioned above. Provide justification for each label. 1. *"I hate those people, they should be banned from this country."* 2. *"Your opinion is stupid, stop talking nonsense."* 3. *"The weather is nice today."*	8
	(b)	Inter-Annotator Agreement (IAA) measures the consistency between different annotators when labeling data. Two common measures are Cohen's Kappa and Krippendorff's Alpha. a) Explain why simple percentage agreement is not enough to measure annotation quality. How do Cohen's Kappa and Krippendorff's Alpha improve on it? b) Two annotators label 50 tweets as either *Hate Speech (H)* or *Not Hate Speech (N)*. The results are as follows: Annotator B: H Annotator B: N	7

	B: H	B: N
Annotator A: H	18	7
Annotator A: N	5	20

Compute the Observed Agreement (Po), Expected Agreement (Pe) and Cohen's Kappa (κ).

c) Why is Krippendorff's Alpha considered more general than Cohen's Kappa? Provide an example of a situation where Krippendorff's Alpha is preferred.

3. (a) Compare and contrast different feature representation techniques in NLP: Bag-of-Words, TF-IDF, Word Embeddings (Word2Vec, GloVe), and contextual embeddings (BERT, GPT). In which scenarios would each technique be most effective? Provide an example of a task where simple Bag-of-Words might outperform embeddings. **6**

(b) You have two documents: **6**

Doc1: "The cat sat on the mat."
Doc2: "The dog sat on the log."
i) Create a vocabulary from these two documents.
ii) Compute the TF (term frequency) of each word in Doc1.
iii) Explain why TF-IDF is useful compared to simple term frequency.

(c) **3**

A student writes the following review:
"The movie was okay, not bad, but could have been better."

i) Identify positive, negative, and neutral phrases.
ii) Suggest a simple rule-based method to assign overall sentiment.
iii) Explain why this sentence might be misclassified by a bag-of-words model.

4. (a) Why have Machine Learning (ML), Deep Learning (DL), and Transformer-based models been successively adopted in NLP? Briefly analyze one key improvement each offers over the previous method. **3**

(b) **5**

Consider the sentence:
"Mr. Maxime visited Germany in 2016 and met Angela Merkel."

a) Tokenize the sentence into words.
b) Assign Part-of-Speech (POS) tags to each token.
c) Identify all proper nouns and explain why POS tagging is useful for Named Entity Recognition (NER).

(c) You are using pre-trained word embeddings. Consider the following words and their cosine similarities: **4**

- Similarity("king", "queen") = 0.78
- Similarity("king", "man") = 0.71
- Similarity("king", "apple") = 0.12

i) Which pair of words is most semantically related? Explain.
ii) Using analogy, "king – man + woman = ?" Which word is expected and why?

	(d)	Named Entity Recognition (NER) systems often struggle with domain-specific or informal text (e.g., social media, medical reports). i) What challenges arise in such scenarios? ii) How can data augmentation, transfer learning, or rule-based methods improve NER performance? ii) Provide a short example illustrating a misclassification and propose a solution.	3
5.	(a)	How does Machine Learning improve over rule-based systems in Natural Language Processing? Justify with an example such as sentiment analysis or POS tagging.	3
	(b)	Compare traditional sequence models (RNN, LSTM) with transformer-based models (BERT, GPT) for NLP tasks. i) Discuss their strengths and weaknesses in handling long text sequences. ii) Give examples of tasks where transformers outperform RNNs, and tasks where RNNs might still be preferable.	4
	(c)	Evaluation of NLP models depends on the task (e.g., text classification, summarization, machine translation). i) Discuss why accuracy, precision, recall, F1-score are used for different tasks. ii) Give examples of scenarios where a single metric might be misleading.	3
	(d)	Machine translation and text summarization systems are evaluated using metrics such as BLEU (Bilingual Evaluation Understudy) and ROUGE (Recall-Oriented Understudy for Gisting Evaluation). i) Explain the main difference between BLEU and ROUGE in terms of precision and recall orientation. ii) The reference translation is: *"The cat is on the mat."* The machine translation output is: *"The cat is on mat."* Compute the BLEU score with unigram and bigram precision (assume no smoothing and equal weights for unigrams and bigrams). iii) For a text summarization task, the reference summary is: *"Global warming affects climate and causes extreme weather."* The system-generated summary is: *"Climate change causes extreme weather patterns."* Explain how ROUGE-1 and ROUGE-2 would evaluate this summary.	5
6.	(a)	Explain unsupervised language discovery and its applications.	5
	(b)	Discuss topic modeling and its use in analyzing language in social networks.	5
	(c)	Describe evolutionary models of language learning and their significance in NLP research.	5

Rangamati Science and Technology University
Department of Computer Science and Engineering

4th Year 1st Semester B.Sc. (Engg.) Final Exam-2023

Course Code: CSE 4106; **Session:** 2019-2020

Course Title: Financial, Cost and Managerial Accounting

Time: 3 Hours **Marks:** 60

NB:
1. Answer any **FOUR (4)** questions out of **SIX (6)** questions.
2. Figures in the right margin indicate marks (15*4=60).
3. All parts of a question must be answered serially.

1. (a) Define accounting. Explain the basic activities of the accounting process with examples. **5**

(b) "An accountant prepares information for different users, some of whom work within the organization and some of whom stay out of the organization." Do you agree with this statement? Illustrate your answer. **5**

(c) Discuss the monetary unit assumption and economic entity assumption. **5**

2. (a) Define Journal, Ledger, Journalizing, and Posting. Explain the technique of journalizing with an example. **5**

(b) What are the steps in the recording process? Explain with examples. **5**

(c) What do you mean by trial balance? Demonstrate the limitations of a trial balance. **5**

3. Mr. Udoy started his own consulting firm, Matrix Consulting, on May 1, 2023. The following transactions occurred during the month of May. **7+8**

1 Mr. Udoy invested $10,000 cash in the business.
2 Paid $125 to advertise in the County News.
3 Purchased $600 of supplies on account.
4 Paid $900 for office rent for the month.
5 Received $4,000 cash for services performed.
6 Withdrew $1,500 cash for personal use.
7 Performed $5,400 of services on account.
8 Paid $2,500 for employee salaries.
9 Paid for the supplies purchased on account in transaction (3).
10 Received a cash payment of $4,000 for services performed on account in transaction (7).
11 Borrowed $5,000 from the bank on a note payable.
12 Purchased equipment for $4,200 on account.
13 Paid $275 for utilities.

Instructions

(a) Prepare a tabular summary of the transactions.
(b) Prepare the income statement, owner's equity statement, and balance sheet at May 31, 2023, for Matrix Consulting.

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4. (a) Define adjusting entries in short. Briefly discuss major types of adjusting entries. 2+3

(b) Explain why adjusting entries are necessary in the accounting cycle. 5

(c) The ACI Company began operations on April 1. At April 30, the trial balance shows the following balances for selected accounts. 5

Prepaid Insurance	3,600
Equipment	28,000
Notes Payable	20,000
Unearned Service Revenue	4,200
Service Revenue	1,800

Analysis reveals the following additional data.

1. Prepaid insurance is the cost of a 2-year insurance policy, effective April 1.
2. Depreciation on the equipment is 500 per month.
3. The note payable is dated April 1. It is a 6-month, 12% notes.
4. Seven customers paid for the company's 6-month lawn service package of 600 beginning in April. The company performed services for these customers in April.
5. ACI Company performed for other customers but not recorded at April 30 totaled 1,500.

Instructions

Prepare the adjusting entries for the month of April. Show computations.

5. (a) What are the three major elements of product costs in a manufacturing company? Explain with examples. 5

(b) What is Cost-Volume-Profit (CVP) analysis? Discuss why companies need CVP analysis in their decision-making. 5

(c) Rondo Corporation's comparative balance sheets are presented below. 5

RONDO CORPORATION

Balance Sheets December 31

	2017	2016
Cash	5,300	3,700
Accounts receivable	21,200	23,400
Inventory	9,000	7,000
Land	20,000	26,000
Buildings	70,000	70,000
Accumulated depreciation—buildings	(15,000)	(10,000)
Total	$1,10,500	$1,20,100
Accounts payable	10,370	31,100
Common stock	75,000	69,000
Retained earnings	25,130	20,000
Total	$1,10,500	$1,20,100

Rondo's 2017 income statement included net sales of $120,000, cost of goods sold of $70,000, and net income of $14,000.

Instructions

Compute the following ratios for 2017.

(i) Current ratio.

(ii) Acid-test ratio.

(iii) Accounts receivable turnover.

(iv) Inventory turnover.

(v) Profit margin.

6. Emily Valley is a licensed dentist. During the first month of the operation of her business, the following events and transactions occurred. **5+6+4**

April 1 Invested $25,000 cash in her business.

1 Hired a secretary-receptionist at a salary of $800 per month.

2 Paid office rent for the month $1,100.

3 Purchased dental supplies on account from Dazzle Company $4,000.

10 Performed dental services and billed insurance companies $5,100.

11 Received $1,000 cash advance from Leah Mataruka for an implant.

20 Received $2,100 cash for services performed from Michael Santos.

25 Withdrew $5000 cash for personal use.

30 Paid secretary-receptionist for the month $2,800.

30 Paid $2,400 to Dazzle for accounts payable due.

Instructions

(a) Journalize the transactions.

(b) Post to the ledger accounts.

(c) Prepare a trial balance on April 30, 2023.